Cap 8 - Inequações

Exemplos

Ex 8.1 a) $3 - x < 5 + 3x$ ⇒ Ineq de 1º grau estritamente menor

b) $5x - 3 \leq x + 15$ ⇒ Ineq de 1º grau menor ou igual

c) $-7 \leq -3x - 3 < 2$ ⇒ Ineq de 1º grau dupla

d) $5 \leq 3x - 1 \leq 14$ ⇒ Ineq de 1º grau dupla

e) $(x - 5)(x + 1) \geq 0$ ⇒ Ineq produto

f) $x^2 - 1 < 0$ ⇒ Ineq do 2º grau

g) $\frac{x+2}{3x-6} \geq 0$ ⇒ Ineq quociente

h) $|x - 2| \leq 7$ ⇒ Ineq modular

i) $|2x| \leq 2|4 - x|$ ⇒ Ineq modular

Ex 8.2 $3 - x < 5 + 3x$

O resultado é arbitrário. Solução geral

$$3 - x < 5 + 3x$$
$$-2 < 4x$$
$$x > -\frac{1}{2} \Rightarrow \boxed{S = \left\{ x \in \mathbb{R} / x > -\frac{1}{2} \right\}}$$

Ex 8.3 Vide Exemplo 7.2

Ex 8.4 a) $3x - 6 \leq 0 \Rightarrow x \leq 2$

$$\boxed{S = \{ x \in \mathbb{R} / x \leq 2 \}}$$

b) $5x - 3 \leq x + 15 \Rightarrow 4x \leq 18 \Rightarrow x \leq \frac{9}{2}$

$$\boxed{S = \left\{ x \in \mathbb{R} / x \leq \frac{9}{2} \right\}}$$

c) $-7 \leq -3x - 3 < 2 \quad (\cdot -1)$
$7 \geq 3x + 3 > -2$

$$-\frac{5}{3} < x \leq \frac{4}{3} \qquad \boxed{S = \left\{ x \in \mathbb{R} / -\frac{5}{3} < x \leq \frac{4}{3} \right\}}$$

d) $6 \leq 3x - 1 \leq 14$

$6 \leq 3x \leq 15$

$2 \leq x \leq 5$

$$S = \{x \in \mathbb{R} \mid 2 \leq x \leq 5\}$$

Ex 8.5 $x^2 + 3x - 10 \leq 0$

(i) Raízes: $\begin{cases} a = 1 \\ b = 3 \\ c = -10 \end{cases}$ $\quad \dfrac{-3 \pm \sqrt{9 + 40}}{2}$

$x' = -5$

$x'' = 2$

(ii) Análise de sinal:

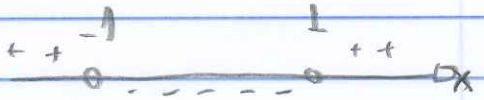


$$S = \{x \in \mathbb{R} \mid -5 \leq x \leq 2\}$$

Ex 8.6 a) $x^2 - 1 < 0$

(i) Raízes: $x^2 - 1 = 0 \Rightarrow x = \pm\sqrt{1} \Rightarrow x' = 1$, $x'' = -1$

(ii) Análise de sinal:



$$S = \{x \in \mathbb{R} \mid -1 < x < 1\}$$

(04)

b) $-x^2 + 2x - 3 \geq 0$

(i) Raízes: $\begin{cases} a = -1 \\ b = 2 \\ c = -3 \end{cases}$ $\quad \dfrac{-2 \pm \sqrt{4 - 12}}{-2}$

$\nexists$ raízes reais

(ii) Análise de sinal:



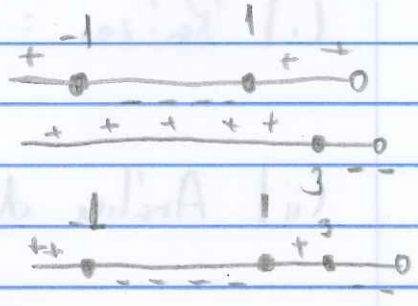
$S = \{\ \}$

Ex 8.7 Apenas explicativo

Ex 8.8 $\underbrace{(x^2 - 1)}_{f(x)} \cdot \underbrace{(-x + 3)}_{g(x)} \geq 0$

(i) Raízes: $f(x) \Rightarrow x' = 1$, $x'' = -1$ $\quad g(x) \Rightarrow x' = 3$

(ii) Análise de sinal: $f(x)$, $g(x)$, $f(x) \cdot g(x)$



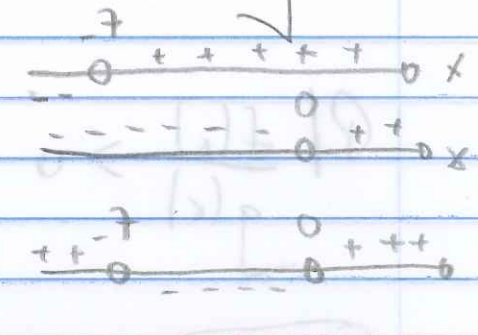
Como queremos $f(x) \cdot g(x) \geq 0$

$S = (-\infty, -1] \cup [1, 3]$

Ex 8.9 $\underbrace{(x+7)}_{f(x)} \cdot \underbrace{(3x)}_{g(x)} > 0$

(i) Raízes: $f(x) \Rightarrow x' = -7$ $\quad g(x) = 0 \Rightarrow x' = 0$

(ii) Análise de sinal:



P/ $f(x).g(x) > 0$: $\boxed{S = (-\infty, -7) \cup (0, \infty)}$

Ex 8.10 Apenas explicativo

Ex 8.11 $\dfrac{\overbrace{x^2 - 3x + 2}^{f(x)}}{\underbrace{x - 2}_{g(x)}} > 0$

(i) Raízes:

$f(x) = \begin{cases} a = 1 \\ b = -3 \\ c = 2 \end{cases}$ $\quad \dfrac{3 \pm \sqrt{9-8}}{2} \Rightarrow \begin{matrix} x' = 2 \\ x'' = 1 \end{matrix}$

$g(x) = 0 \quad x' = 2$

(i) Análise de sinal: $f(x)$



$g(x)$



$\frac{f(x)}{g(x)}$



P/ $\frac{f(x)}{g(x)} > 0 \Rightarrow$ $S = \{x \in \mathbb{R} / x > 1\}$

Ex 8.12 $\frac{x+2}{3x-6} \leq 0$

$f(x) = x+2$

$g(x) = 3x-6$

(i) Raízes: $f(x) \Rightarrow x' = -2$

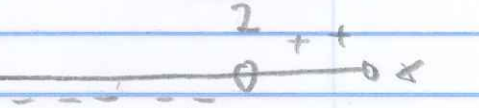
$g(x) \Rightarrow x' = 2$

(ii) Análise de sinal:

$f(x)$



$g(x)$



$\frac{f(x)}{g(x)}$



Logo, P/ $\frac{f(x)}{g(x)} \leq 0 \Rightarrow$ $S = \{x \in \mathbb{R} / -2 \leq x < 2\}$

Ex 8.13 Apenas explicativo

Ex 8.14 a) $|5-x| \geq 0$

Como, pela definição de módulo $|5-x|$ sempre será positivo, então

$$S = \mathbb{R}$$

b) $|3x+2| < 1$. Pela definição de módulo

$$-1 < 3x + 2 < 1$$
$$-3 < 3x < -1$$
$$-1 < x < -\frac{1}{3}$$

$$S = \left\{ x \in \mathbb{R} \mid -1 < x < -\frac{1}{3} \right\}$$

Lista de Exercícios

Inequações

01 a) $3x + 8 > 0 \Rightarrow x > -\frac{8}{3}$

$$\boxed{S = \left\{ x \in \mathbb{R} / x > -\frac{8}{3} \right\}}$$

b) $-2x + 4 \leq 3 \quad (-1) \Rightarrow 2x - 4 \geq -3$

$$x \geq 1/2$$

$$\boxed{S = \left\{ x \in \mathbb{R} / x \geq \frac{1}{2} \right\}}$$

c) $(x-2)^2 < (2x-1) \Rightarrow x^2 - 4x + 4 - 2x + 1 < 0$

$$x^2 - 6x + 5 < 0$$

(i) Raízes: $\begin{cases} a = 1 \\ b = -6 \\ c = 5 \end{cases} \quad \frac{6 \pm \sqrt{36 - 20}}{2} \Rightarrow \begin{matrix} x' = 5 \\ x'' = 1 \end{matrix}$

c) Análise de sinal : $\overset{+\ +}{\underset{}{}}\ 1\ \overset{-\ -\ -}{\underset{}{}}\ 5\ \overset{+\ +}{\underset{}{}}$ x

$$\boxed{S = \{x \in \mathbb{R} / 1 < x < 5\}}$$

d) $3 - 2x \geq 9 + 4x \Rightarrow -6 \geq 6x \Rightarrow x \leq -1$

$$\boxed{S = \{x \in \mathbb{R} / x \leq -1\}}$$

e) $2x + 7 > 3 \Rightarrow 2x > -4 \Rightarrow x > -2$

$$\boxed{S = \{x \in \mathbb{R} / x > -2\}}$$

f) $\frac{1}{5} - \frac{x}{2} \geq \frac{7}{10} \Rightarrow \frac{2 - 5x}{10} \geq \frac{7}{10}$

$2 - 5x \geq 7 \Rightarrow -5 \geq 5x \Rightarrow x \leq -1$

$$\boxed{S = \{x \in \mathbb{R} / x \leq -1\}}$$

g) $\frac{1}{x+1} < \frac{2}{3x-1} \Rightarrow \frac{1}{x+1} - \frac{2}{3x-1} < 0$

$$\frac{3x-1-2x-2}{(x+1)(3x-1)} < 0 \Rightarrow \frac{x-3}{3x^2+2x-1} < 0$$

(i) Raízes: $g(x) \Rightarrow \begin{cases} a = 3 \\ b = 2 \\ c = -1 \end{cases}$

$$\frac{-2 \pm \sqrt{4+12}}{6} \Rightarrow \begin{matrix} x' = -1 \\ x'' = \frac{1}{3} \end{matrix}$$

$f(x) \Rightarrow x' = 3$

(ii) Análise de sinal:

$f(x)$ — bolinha em 3, + + + à direita

$g(x)$ — bolinhas em -1 e $1/3$, + + + + à direita

$\frac{f(x)}{g(x)}$ — bolinhas em -1, $1/3$ e 3, + + + à direita

P/ $\frac{f(x)}{g(x)} < 0 \Rightarrow$

$$S = (-\infty, -1) \cup (1/3, 3)$$

(04)

h) $\frac{x+2}{3} + \frac{1+2x}{2} < \frac{13}{6} + x$

$$\frac{x+2}{3} + \frac{1+2x}{2} - \frac{13}{6} < 0$$

$$\frac{2x+4+3+6x-13-6x}{6} < 0$$

$$2x - 6 < 0 \Rightarrow x < 3$$

$$\boxed{S = \{x \in \mathbb{R} / x < 3\}}$$

i) $3 - 2x \leq 3x - 1 \leq 5$

Resolvendo cada lado em separado

(.) $3 - 2x \leq 3x - 1 \Rightarrow x \geq \frac{4}{5} \Rightarrow$ 

(..) $3x - 1 \leq 5 \Rightarrow x \leq 2 \Rightarrow$ 

$$\boxed{S = \left\{x \in \mathbb{R} / \frac{4}{5} \leq x \leq 2\right\}}$$

j) $x^2 - 3x + 2 > 0$

Raízes: $\begin{cases} a = 1 \\ b = -3 \\ c = 2 \end{cases}$ $\quad \dfrac{3 \pm \sqrt{9 - 8}}{2} \Rightarrow \begin{matrix} x' = 2 \\ x'' = 1 \end{matrix}$

Análise de sinal:

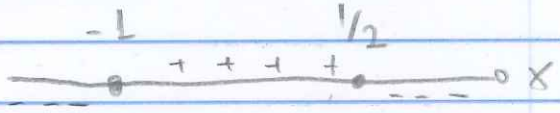


$$S = \{x \in \mathbb{R} / x < 1 \text{ ou } x > 2\}$$

k) $-2x^2 - x + 1 \geq 0$

Raízes: $\begin{cases} a = -2 \\ b = -1 \\ c = 1 \end{cases}$ $\quad \dfrac{1 \pm \sqrt{1 + 8}}{-4} \Rightarrow \begin{matrix} x' = -1 \\ x'' = \dfrac{1}{2} \end{matrix}$

Análise de sinal:



$$S = \left\{x \in \mathbb{R} / -1 \leq x \leq \frac{1}{2}\right\}$$

06

c) $x^2 - 7x + 10 \leq 0$

Raízes: $\begin{cases} a = 1 \\ b = -7 \\ c = 10 \end{cases}$ $\quad \dfrac{7 \pm \sqrt{49 - 40}}{2} \quad \begin{matrix} x' = 2 \\ x'' = 5 \end{matrix}$

Análise de sinal:



$$S = \{x \in \mathbb{R} \mid 2 \leq x \leq 5\}$$

m) $-5x^2 - 14x + 3 \geq 0$

Raízes: $\begin{cases} a = -5 \\ b = -14 \\ c = 3 \end{cases}$ $\quad \dfrac{14 \pm \sqrt{196 + 60}}{-10} \Rightarrow \dfrac{14 \pm \sqrt{256}}{-10}$

$x' = -3$

$x'' = \dfrac{1}{5}$

Análise de sinal:



$$S = \{x \in \mathbb{R} \mid -3 \leq x \leq 1/5\}$$

n) $x^2 - 6x + 9 \leq 0$

Raízes: $\begin{cases} a = 1 \\ b = -6 \\ c = 9 \end{cases}$ $\quad \frac{6 \pm \sqrt{36 - 36}}{2} \Rightarrow x' = x'' = 3$

Análise de sinal: $\quad + + + \; 3 \; + + +$

$$\boxed{S = \{x \in \mathbb{R} / \; x = 3\}}$$

o) $\frac{2x}{3} + 3 < \frac{x^2 + 2x}{3} \Rightarrow \frac{2x}{3} + \frac{9}{3} - \frac{(x^2 + 2x)}{3} < 0$

$$2x + 9 - x^2 - 2x < 0$$

$$-x^2 + 9 < 0$$

Raízes: $\quad -x^2 + 9 = 0 \Rightarrow x = \pm\sqrt{9} \Rightarrow x' = 3$

$x'' = -3$

Análise de sinal: $\quad - - \; -3 \; + + + \; 3 \; - -$

$$\boxed{S = \{x \in \mathbb{R} / \; x < -3 \text{ ou } x > 3\}}$$

(a)

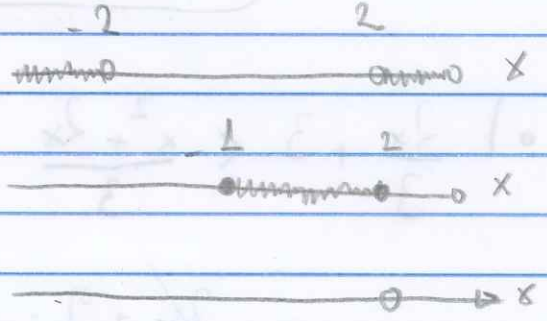
f) $2 < x^2 - 2 \leq x$ (Resolver cada lado)

* $x^2 - 2 > 2 \Rightarrow x^2 - 4 > 0 \Rightarrow x' = 2$, $x'' = -2$

** $x^2 - 2 \leq x \Rightarrow x^2 - x - 2 \leq 0$

$\begin{cases} a = 1 \\ b = -1 \\ c = -2 \end{cases}$ $\quad \frac{1 \pm \sqrt{1+8}}{2} \Rightarrow x' = 2$, $x'' = -1$

Análise de sinal:



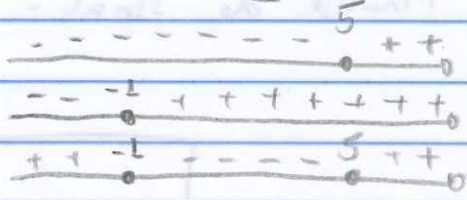
Não há intersecção, logo: $\boxed{S = \{\ \}}$

g) $\underbrace{(x-5)}_{f(x)} \cdot \underbrace{(x+1)}_{g(x)} \geq 0$

$f(x) \Rightarrow x' = 5$

$g(x) \Rightarrow x' = -1$

Análise de sinal:



$\boxed{S = \{x \in \mathbb{R} \mid x \leq -1 \text{ ou } x \geq 5\}}$

f(x) g(x)

c) $\underbrace{(x^2+7x)}_{f(x)} \cdot \underbrace{(3-x)}_{g(x)} \leq 0$

Raízes: $f(x) \Rightarrow x(x+7) = 0 \Rightarrow x' = 0$

$x'' = -7$

$g(x) \Rightarrow x = 3$

Análise de sinal: $f(x)$



$g(x)$



$f(x) \cdot g(x)$



$$S = \{x \in \mathbb{R} / -7 \leq x \leq 0 \text{ ou } x \geq 3\}$$

e) $(x-3)\cdot(x^2-6x+5) < 0$

Raízes: $f(x) \Rightarrow x = 3$

$g(x) \Rightarrow \frac{6 \pm \sqrt{36-20}}{2} \Rightarrow x' = 5$

$x'' = 1$

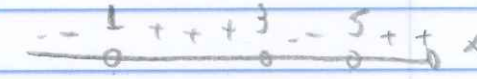
Análise de sinal: $f(x)$



$g(x)$



$f(x) \cdot g(x)$



$$S = \{x \in \mathbb{R} / x < 1 \text{ ou } 3 < x < 5\}$$

(10)

t) $(x+7)(3-x^2) \leq 0$

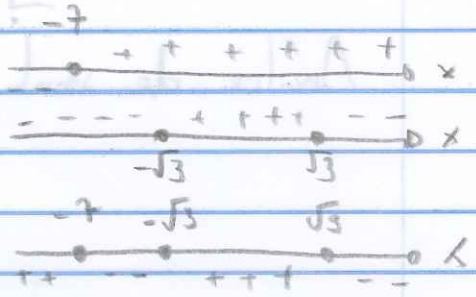
Racines : $f(x) \Rightarrow x' = -7$

$g(x) \Rightarrow x' = \sqrt{3}$
$x'' = -\sqrt{3}$

Analyse de signe : $f(x)$

$g(x)$

$f(x) \cdot g(x)$



$$S = \{x \in \mathbb{R} / -7 \leq x \leq -\sqrt{3} \text{ ou } x \geq \sqrt{3}\}$$

u) $(x^2 - 2x - 8)(1-x) \leq 0$

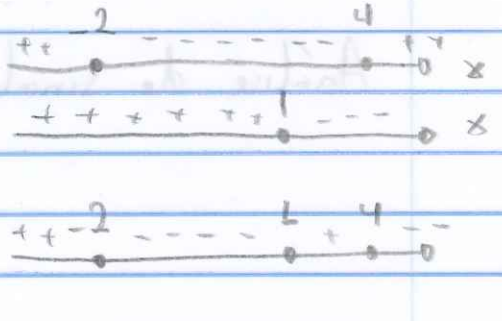
Racines : $f(x) \Rightarrow \frac{2 \pm \sqrt{4+32}}{2} \Rightarrow x' = 4$
$x'' = -2$

$g(x) \Rightarrow x' = 1$

Analyse de signe : $f(x)$

$g(x)$

$f(x) \cdot g(x)$



$$S = \{x \in \mathbb{R} / -2 \leq x \leq 1 \text{ ou } x \geq 4\}$$

v) $\frac{1+x}{x-4} \geq 0$

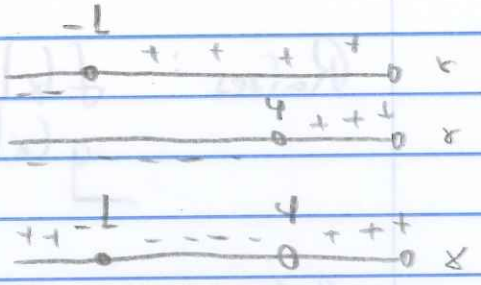
Raízes: $f(x) = 0 \Rightarrow x' = -1$

$g(x) = 0 \Rightarrow x' = 4$

Análise de sinal: $f(x)$

$g(x)$

$f(x)/g(x)$



$$S = \{x \in \mathbb{R} \mid x \leq -1 \text{ ou } x > 4\}$$

vi) $\frac{2x-3}{x-1} - 5 > 0 \Rightarrow \frac{2x-3-5x+5}{x-1} > 0$

$$\frac{-3x+2}{x-1} > 0$$

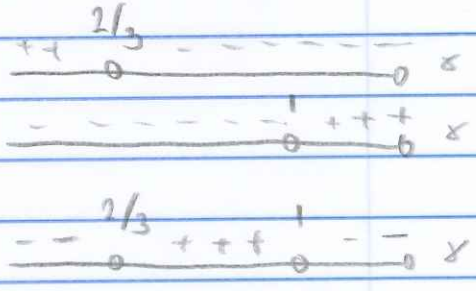
Raízes: $f(x) = 0 \Rightarrow x' = 2/3$

$g(x) = 0 \Rightarrow x' = 1$

Análise de sinal: $f(x)$

$g(x)$

$f(x)/g(x)$



$$S = \{x \in \mathbb{R} \mid 2/3 < x < 1\}$$

(12)

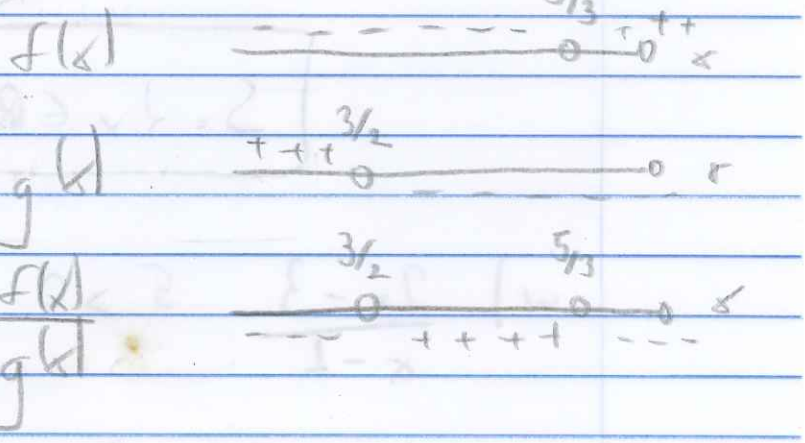
$$\frac{x-2}{-2x+3} - 1 > 0 \Rightarrow \frac{x-2+2x-3}{-2x+3} > 0$$

$$\frac{3x-5}{-2x+3} > 0$$

Raízes: $f(x) \Rightarrow x' = 5/3$

$g(x) \Rightarrow x' = 3/2$

Análise de sinal:



$$S = \left\{ x \in \mathbb{R} \mid \frac{3}{2} < x < \frac{5}{3} \right\}$$

02) a) $|x+4| < 7$

Elevando ambos ao quadrado: $(x+4)^2 < 7^2 \rightarrow x^2 + 8x + 16 < 49$

$x^2 + 8x - 33 < 0$

Encontramos as raízes, por Bhaskara:

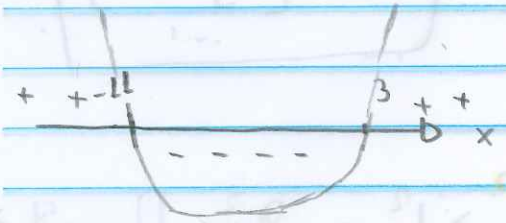
$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$\begin{cases} a = 1 \\ b = 8 \\ c = -33 \end{cases}$

$\frac{-8 \pm \sqrt{64 + 132}}{2} \rightarrow \frac{-8 \pm 14}{2} \rightarrow x' = \frac{-8-14}{2} = -11$

$x'' = \frac{-8+14}{2} = 3$

Fazendo o estudo do sinal:



Como a condição é < 0 (Negativo), então

$S = (-11, 3)$

Obs.: O intervalo é aberto pois = 0 está fora da condição

b) $|2x - 5| < 3 \rightarrow (2x-5)^2 < 3^2 \rightarrow 4x^2 - 20x + 25 < 9$

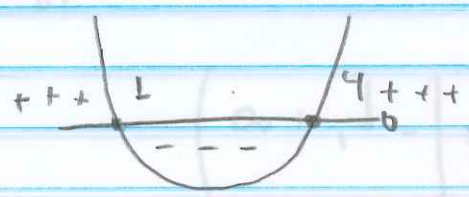
$\begin{cases} a = 4 \\ b = -20 \\ c = 16 \end{cases}$

$\frac{20 \pm \sqrt{400 - 256}}{8}$

$4x^2 - 20x + 16 < 0$

$x' = \frac{20+12}{8} = 4$

$x'' = \frac{20-12}{8} = 1$



$S = (1, 4)$

credeal

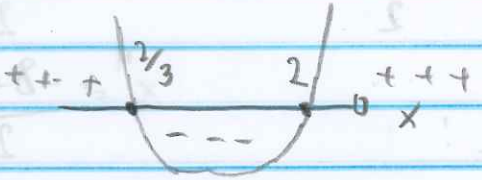
(14)

c) $|3x-4| \leq 2 \Rightarrow (3x-4)^2 \leq 2^2 \Rightarrow 9x^2 - 24x + 16 \leq 4$

$\begin{cases} a = 9 \\ b = -28 \\ c = 12 \end{cases}$ $\quad \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ $\quad 9x^2 - 24x + 12 \leq 0$

$\dfrac{24 \pm \sqrt{576 - 432}}{18} \Rightarrow \dfrac{24 \pm \sqrt{144}}{18} \Rightarrow x' = \dfrac{24+12}{18} = 2$

Analisando o sinal: $\quad x'' = \dfrac{12}{18} = \dfrac{2}{3}$



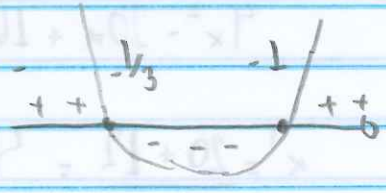
$S = \left[\dfrac{2}{3}, 2\right]$

Obs.: O Intervalo é fechado pois a condição é ≤ 0

d) $|3x+2| \geqslant 1 \Rightarrow (3x+2)^2 \geqslant 1^2 \Rightarrow 9x^2 + 12x + 4 \geq 1$

$\begin{cases} a = 9 \\ b = 12 \\ c = 3 \end{cases}$ $\quad \dfrac{-12 \pm \sqrt{144 - 108}}{18}$ $\quad 9x^2 + 12x + 3 \geq 0$

$x' = \dfrac{-12+6}{18} \Rightarrow \dfrac{-6}{18} \Rightarrow -\dfrac{1}{3}$



$x'' = \dfrac{-12-6}{18} \Rightarrow \dfrac{-18}{18} \Rightarrow -1$

$S = \left(-\infty, -\dfrac{1}{3}\right] \cup \left[-1, +\infty\right)$

credeal

e) $|5-x| > 7 \Rightarrow (5-x)^2 > 7^2 \Rightarrow 25 - 10x + x^2 > 49$

$\begin{cases} a = 1 \\ b = -10 \\ c = -24 \end{cases}$ $\quad \dfrac{10 \pm \sqrt{100+96}}{2}$ $\quad x^2 - 10x - 24 > 0$

$x' = \dfrac{10+14}{2} = \dfrac{24}{2} = 12$



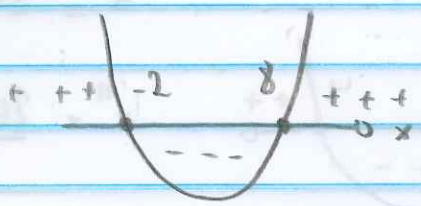
$x'' = \dfrac{10-14}{2} \Rightarrow \dfrac{-4}{2} = -2$

$S = (-\infty, -2) \cup (12, +\infty)$

f) $|3-x| < 5 \Rightarrow (3-x)^2 < 5^2 \Rightarrow 9 - 6x + x^2 < 25$

$\begin{cases} a = 1 \\ b = -6 \\ c = -16 \end{cases}$ $\quad \dfrac{6 \pm \sqrt{36+64}}{2}$ $\quad x^2 - 6x - 16 < 0$

$x' = \dfrac{6+10}{2} = 8$



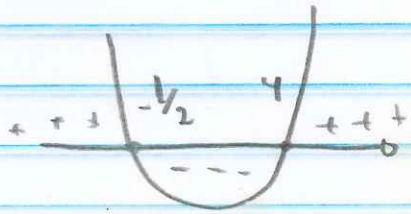
$x'' = \dfrac{6-10}{2} \Rightarrow -2$

$S = (-2, 8)$

g) $|7-4x| \leq 9 \Rightarrow (7-4x)^2 \leq 9^2 \Rightarrow 49 - 56x + 16x^2 \leq 81$

$\begin{cases} a = 16 \\ b = -56 \\ c = -32 \end{cases}$ $\quad \dfrac{56 \pm \sqrt{3136+2048}}{32}$ $\quad 16x^2 - 56x - 32 \leq 0$

$x' = \dfrac{56+72}{32} = 4$



$x'' = \dfrac{56-72}{32} = -\dfrac{16}{32} = -\dfrac{1}{2}$

$S = \left[-\frac{1}{2}, 4\right]$

credeal

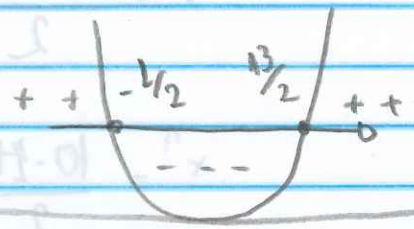
(16)

h) $|6-2x| \geq 7 \Rightarrow (6-2x)^2 \geq 7^2 \Rightarrow 36-24x+4x^2 \geq 49$

$\begin{cases} a=4 \\ b=-24 \\ c=-13 \end{cases}$ $\quad \dfrac{24 \pm \sqrt{576+208}}{8}$ $\quad 4x^2-24x-13 \geq 0$

$x' = \dfrac{24+28}{8} = \dfrac{52}{8} = \dfrac{13}{2}$

$x'' = \dfrac{24-28}{8} = -\dfrac{4}{8} = -\dfrac{1}{2}$



$S = \left(-\infty, -\dfrac{1}{2}\right] \cup \left[\dfrac{13}{2}, +\infty\right)$

i) $|2x-5| > 3 \Rightarrow (2x-5)^2 > 3^2 \Rightarrow 4x^2-20x+25 > 9$

$\begin{cases} a=4 \\ b=-20 \\ c=16 \end{cases}$ $\quad \dfrac{20 \pm \sqrt{400-256}}{8}$ $\quad 4x^2-20x+16 > 0$

$x' = \dfrac{20+12}{8} = 4$

$x'' = \dfrac{20-12}{8} = \dfrac{8}{8} = 1$



$S = (-\infty, 1] \cup (4, +\infty)$

credeal

j) $|x+4| \leq |2x-6| \Rightarrow (x+4)^2 \leq (2x-6)^2$

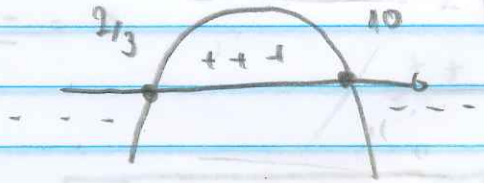
$x^2 + 8x + 16 \leq 4x^2 - 24x + 36$

$-3x^2 + 32x - 20 \leq 0$

$\begin{cases} a = -3 \\ b = 32 \\ c = -20 \end{cases}$ $\quad \dfrac{-32 \pm \sqrt{1024 - 240}}{-6}$

$x' = \dfrac{-32+28}{-6} = \dfrac{-4}{-6} = \dfrac{2}{3}$

$x'' = \dfrac{-32-28}{-6} = \dfrac{60}{6} = 10$



$S = \left(-\infty, \frac{2}{3}\right] \cup [10, +\infty)$

k) $|13x| > |6-3x| \Rightarrow (3x)^2 > (6-3x)^2$

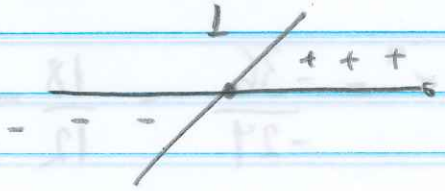
$9x^2 > 36 - 36x + 9x^2$

$36x > 36$

$x > \dfrac{36}{36}$

$x > 1$

$S = (1, +\infty)$



credeal

(18)

l) $|13 + 2x| < |4 - x| \Rightarrow (3+2x)^2 < (4-x)^2$

$9 + 12x + 4x^2 < 16 - 8x + x^2$

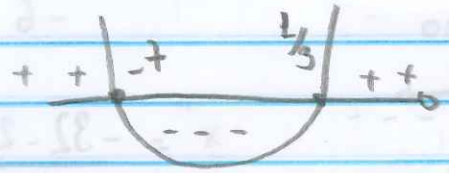
$3x^2 + 20x - 7 < 0$

$\begin{cases} a = 3 \\ b = 20 \\ c = -7 \end{cases}$

$\frac{-20 \pm \sqrt{400 + 84}}{6} \Rightarrow x' = \frac{-20+22}{6} = \frac{2}{6} = \frac{1}{3}$

$x'' = \frac{-20-22}{6} = \frac{-42}{6}$

$x'' = -7$



$S = (-7, \frac{1}{3})$

m) $|9 - 2x| \geq |4x| \Rightarrow (9-2x)^2 \geq (4x)^2$

$\begin{cases} a = -12 \\ b = -36 \\ c = 81 \end{cases}$

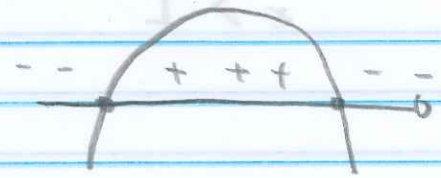
$81 - 36x + 4x^2 \geq 16x^2$

$-12x^2 - 36x + 81 \geq 0$

$\frac{36 \pm \sqrt{1296 + 3888}}{-24} \Rightarrow \frac{36 \pm \sqrt{5184}}{-24} \Rightarrow \frac{36 \pm 72}{-24}$

$x' = \frac{108}{-24} = -\frac{9}{2}$

$x'' = \frac{-36}{-24} = \frac{18}{12} \Rightarrow \frac{3}{2}$



$S = \left[-\frac{9}{2}, \frac{3}{2}\right]$

credeal

a) $|x^2 + x - 1| < 1$

$-1 < x^2 + x - 1 < 1$

Resolvendo cada um dos lados:

* $x^2 + x - 1 > -1$

$x^2 + x > 0$ ⇒ $x' = 0$, $x'' = -1$

** $x^2 + x - 1 < 1$ ⇒ $x^2 + x - 2 < 0$

$\begin{cases} a = 1 \\ b = 1 \\ c = -2 \end{cases}$ $\quad \frac{-1 \pm \sqrt{1 + 8}}{2} \Rightarrow \frac{-1 \pm 3}{2} \Rightarrow$ $x' = -2$, $x'' = 1$

Análise de sinal:

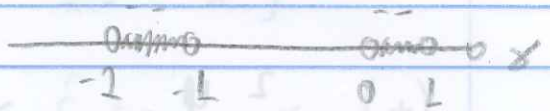
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Interseção

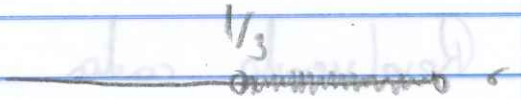


$S = \{x \in \mathbb{R} / -2 < x < -1 \text{ ou } 0 < x < 1\}$

03) a) $\begin{cases} 3x - 1 > 0 & (*) \\ 3x > 5x - 2 & (**) \end{cases}$

$(*)\ 3x - 1 > 0$

$x > \frac{1}{3}$



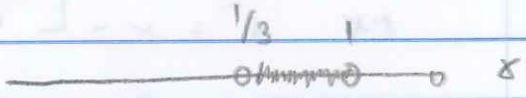
$(**)\ 3x > 5x - 2$

$2x < 2$

$x < 1$



Interseção



$$S = \{x \in \mathbb{R} \mid 1/3 < x < 1\}$$

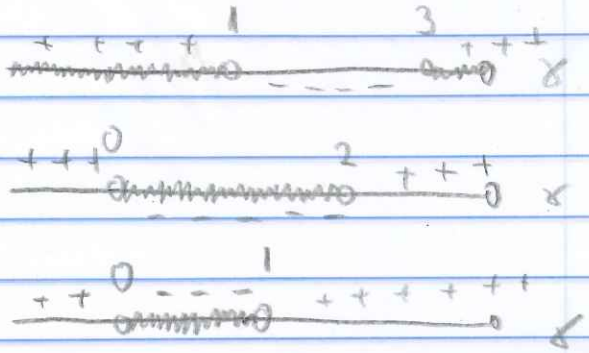
b) $\begin{cases} x^2 - 4x + 3 > 0 & (*) \\ x^2 - 2x < 0 & (**) \end{cases}$

$(*)\ x^2 - 4x + 3 > 0$

$\begin{cases} a = 1 \\ b = -4 \\ c = 3 \end{cases}$ $\quad \frac{4 \pm \sqrt{16 - 12}}{2} \Rightarrow \begin{matrix} x' = 3 \\ x'' = 1 \end{matrix}$

$(**)\ x^2 - 2x < 0 \Rightarrow \begin{matrix} x' = 0 \\ x'' = 2 \end{matrix}$

Análise de sinal:



Interseção

$$S = \{x \in \mathbb{R} / 0 < x < L\}$$